

Chapter 1 – INTRODUCTION TO PROJECT

1.1 ABSTRACT

The Greenhouse effect is a leading factor in keeping the Earth warm because it keeps some of the planet's heat that would otherwise escape from the atmosphere out to space. The study report on the Greenhouse gases and their impact on Global warming. Without the greenhouse effect the Earth's average global temperature would be much colder and life on Earth as we know it would be impossible. Greenhouse gases include water vapor, CO₂, methane, nitrous oxide (N₂O) and other gases. Carbon dioxide (CO₂) and other greenhouse gases turn like a blanket, gripping Infra-Red radiation and preventing it from escaping into outer space. The clear effect of the greenhouse gases is the stable heating of Earth's atmosphere and surface, thus, global warming. The ability of certain gases, greenhouse gases, to be transparent to inbound visible light from the sun, yet opaque to the energy radiated from the earth is one of the best still events in the atmospheric sciences. The existence of greenhouse effect is what makes the earth a comfortable place for life. The study also reveals the importance of greenhouse gases to the warming of the planet earth.

1.2 RATIONALE

A greenhouse is a house made of glass that can be used to grow plants. The sun's radiations warm the plants and the air inside the greenhouse. The heat trapped inside can't escape out and warms the greenhouse which is essential for the growth of the plants. Same is the case in the earth's atmosphere.

During the day the sun heats up the earth's atmosphere. At night, when the earth cools down the heat is radiated back into the atmosphere. During this process, the heat is absorbed by the greenhouse gases in the earth's atmosphere. This is what makes the surface of the earth warmer, that makes the survival of living beings on earth possible.

However, due to the increased levels of greenhouse gases, the temperature of the earth has increased considerably. This has led to several drastic effects.

Let us have a look at the greenhouse gases and understand the causes and consequences of greenhouse effects with the help of a diagram.

1.3 COURSE OUTCOMES ADDRESSED

- i. Develop public awareness about environment.

Chapter 2 – LITERATURE REVIEW

Green House Effect is heating up of earth's atmosphere due to the trapping of intra-red ray. (reflected from the earth's surface) by the carbon dioxide layer in the atmosphere is called green-house effect.

The green-house effect in the atmosphere occurs due to the presence of a blanket of carbon-dioxide gas in the atmosphere. This blanket of carbon dioxide gas in the atmosphere allows the sunlight to come in freely but does not allow the intra-red radiation reflected by the earth's surface to go out. It is just because the sun light can come in freely but the intra-red rays cannot go out freely that the temperature of earth's atmosphere is raised.

The rise in temperature produce gas in the by green-house effect on earth's atmosphere depends on the amount of carbon dioxide gas in the atmosphere. In other words, the proportion of carbon dioxide in atmosphere effects the temperature of atmosphere. So, if the proportion of carbon dioxide gas in the atmosphere increases, than the temperature of earth's atmosphere will also rise further.

Chapter 3 – PROJECT DETAILS

3.1 ACTUAL METHODOLO

(Mention the actual resources used. **DELETE THIS LINE FROM REPORT.**)

Sr. No.	Name of Resource/Material	Specifications	Quantity	Remarks
1.	Computer system	HP Pavilion	1	
2.	software	Microsoft word	1	
3.	Operating system	Windows 10	1	
4.	Primary memory	512 GB	1	
5.	Ram	8 GB ram	1	
6	Processor	AMD Ryzen 5	1	

Chapter 4 – MICRO-PROJECT OUTPUTS

4.1 OUTPUTS OF THE MICRO-PROJECT

Sources of Green-House Effect:

As this keeps on happening, the Earth will become warmer and warmer. This process is also commonly known as the process of 'Global Warming'. The process of green-house effect on earth is quite similar to a green-house. A green-house is a house made out of glass. It is used to grow plants in the winter season.

In the green-house the rays of the sun come in via the glass, and then the heat remains trapped within the glass walls and is not allowed to escape. In this manner the temperature inside the green-house remains high from outside and allows the plant to grow during the winter season as well.

Mainly human beings are responsible for what is happening to the Earth. The production of Green-house Gases is largely due to the burning of fossil fuels such as petroleum, coal and natural gas and also the burning of rain forests. This has served to speed up the Green-house Effect process which happens naturally. This is also, therefore, referred to as the

"Man-Made Green-house Effect".

As mentioned before, the increasing heat which remains trapped within the atmospheric layer will end up creating warmer climates across the globe. This process will significantly cause various climatic disturbances all over the Earth. This can result in many problems in the long run.

To combat this rising threat of global warming, countries all over the world are trying to create pollution free environment and planting trees. Re-forestation is a very natural method of creating an environment where global warming can be controlled as plants and trees actually absorb carbon dioxide and release oxygen into the atmosphere.

Automobile companies are focusing on manufacturing 'Green' cars which release reduced amounts of pollutants into the atmosphere. In the solar system the atmosphere of Venus, Mars and Titan also contains gases that cause green-house effect.

Problem from Green House Effect

The problem is that human activity may be making the green house gas blanket "thicker" For example, burning fossil fuel throws huge amounts of CO₂ into the air, the destruction of forests allows carbon stored in the trees to escape into the atmosphere and other activities such as raising cattle and planting rice emit methane, nitrous oxide, and other green house gases. Until man kind began burning fossil fuels, green house gases that occur naturally remained in relative balance. But the beginning of the Industrial Revolution in Britain ushered in rapid industrialization that greatly increased man's assault on the ecology.

Causes of Green-House Effect:***i. Deforestation:***

One of the man-made causes of the Green-house effect is deforestation. Deforestation increases the amount of carbon-di-oxide in the atmosphere. Also, due to the disappearance of trees, photosynthesis cannot take place.

Deforestation is rampant today due to the increase in human population. The levels of deforestation have increased by about nine per-cent in recent years. Also, the burning of wood causes it to decay, therefore releasing more carbon-dioxide into the atmosphere.

ii. Industrial Emission:

Green-house Gases are also released into the atmosphere due to the burning of fossil fuels, oil, coal and gas. These materials are used increasingly and rampantly in Industries. Therefore Industries are also a major cause of the Green-house Effect. Manufacturing of goods also leads to an increase in the level of green-house gases.

Burning of coal and natural gas to run the factories has led to the dangerous rise of carbon dioxide and methane. This has a radical impact on the green-house effect and contributes to global warming. Since the chimneys are tall, fumes get mixed to the air directly and are responsible for trapping more heat and increasing it.

iii. Electrical Emission:

Another man-made cause of the increase in the Green-house effect is due to the emission of such gases from all electrical appliances. Refrigerator in the house emits gases.

These gases are known as chlorofluorocarbons (CFCs) and are also used in refrigerators, aerosol cans, some foaming agents in the packaging industry, fire extinguisher chemicals and cleaners used in the electronic industry all contribute to the green-house effect. Some processes of the cement manufacturing industries also act as a cause towards the Green-house effect.

iv. Fuel Burning:

Other man-made processes that contribute and are a cause to the Green-house effect are burning of gasoline, oil and coal. Apart from these, some farming and land-use processes are also a cause of the Green-house effect.

Most factories also produce many gases which last for a longer time in the atmosphere. These gases contribute to the green-house effect and also to the global warming on the planet. These gases are not naturally available in the atmosphere.

The burning of fossil fuels like petrol, diesel i.e. emissions from automobiles increase the amount of carbon dioxide in the environmental. Other gases like carbon monoxide and sulphur dioxide are also emitted from the exhaust pipes of vehicles. These gases contribute towards air pollution, which causes a dramatic addition of green-house gases in the atmosphere.

v. Population:

Population growth also is an indirect contributor and one of the causes of the Green-house effect. With the increase in population, the needs and wants of the people increase. Therefore, this increases the manufacturing processes as well as the industry processes.

This results in the increase of the release of industrial gases which catalysed the green-house-effect. The increase in population also results in the increase of agricultural processes. Most man-made machines, like the automobiles also contribute to the green-house effect.

There can be a number of reasons for green-house gases leading to global warming, some pertaining to human activity.

4.2 SKILL DEVELOPED/LEARNING OUT OF THIS MICRO-PROJECT

Greenhouse gases' are crucial to keeping our planet at a suitable temperature for life. Without the natural greenhouse effect, the heat emitted by the Earth would simply pass outwards from the Earth's surface into space and the Earth would have an average temperature of about -20°C.

Chapter 5

ADVANTAGES, DISADVANTAGES & APPLICATIONS

5.1 ADVANTAGES

- GREENHOUSE EFFECT PROMOTES LIFE
- PROTECTION FROM DANGER

5.2 DISADVANTAGES

- GLOBAL WARMING
- RISE IN SEA LEVELS

5.3 APPLICATIONS

- Growing vegetables and flowers in snow-covered areas
- Use of glass to keep homes warm

Chapter 6

CONCLUSIONS, FUTURE SCOPE & REFERENCES

6.1 CONCLUSION

So far not many measures have been taken to address climate change. This is largely caused by the major uncertainties still surrounding the theory. But climate change is also a global problem that is hard to solve by single countries. Therefore in 1998, the Kyoto Protocol was negotiated in Kyoto, Japan. It requires participating countries to reduce their anthropogenic greenhouse gas emissions (CO₂, CH₄, N₂O, HFCs, PFCs, and SF₆) by at least 5% below 1990 levels in the commitment period 2008 to 2012. The Kyoto Protocol was eventually signed in Bonn in 2001 by 186 countries. Several countries such as the United States and Australia have retreated. From 1998 onwards the terminology on the greenhouse effect started to change as a result of media influences. The greenhouse effect as a term was used fewer and fewer and people started to refer to the theory as either global warming or climate change.

6.2 AREA OF FUTURE IMPROVEMENT

As this keeps on happening, the Earth will become warmer and warmer. This process is also commonly known as the process of 'Global Warming'. The process of green-house effect on earth is quite similar to a green-house. A green-house is a house made out of glass. It is used to grow plants in the winter season.

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6.3 REFERENCES

<https://www.environmentalpollution.in/project-report/project-report-on-green-house-effect/1613>
<https://www.conserve-energy-future.com/greenhouseeffectcauses.php>

WEEKLY WORK / PROGRESS REPORT

Details of 16 Engagement Hours of the Student Regarding Completion of the Project						
Week No.	Date	Timing			Work/Activity Performed	Sign of Guide
		From	To	Duration in Hours		
1.	02-08-23	3 pm	5 pm	2 hours	Project Selection Discussion of various Project Ideas, Project Topic Selection & Group Nomination	
2.	09-08-23	3 pm	5 pm	2 hours	Problem Identification & Concept Discussion Discussion of Project scope, Abstract Writing, Decide Modules, etc.	
3.	16-08-23	3 pm	5 pm	2 hours	Project Planning & Scheduling Team structure & Work Distribution	
4.	23-08-23	3 pm	5 pm	2 hours	Project Concept Demonstration Prepare a Presentation & present the same	
5.	30-08-23	3 pm	5 pm	2 hours	Literature Review Project related data collection & study of similar existing system with their Drawbacks, etc	
6.	06-09-23	3 pm	5 pm	2 hours	Requirement Analysis Minimum Hardware & Software Requirement	
7.	13-09-23	3 pm	5 pm	2 hours	Design-I Algorithm design, Flow design	
8.	20-09-23	3 pm	5 pm	2 hours	Design-II Form/View design - as per the requirement of the various proposed modules of project Coding-I Front End Start coding GUI of various Forms / Modules	
9.	27-09-23	3 pm	5 pm	2 hours	Coding-I Front End Start coding GUI of various Forms / Modules	
10.	03-10-23	3 pm	5 pm	2 hours	Coding-II Back End Coding of Data Flow, Database, Connect the various forms	
11.	17-10-23	3 pm	5 pm	2 hours	Testing Preparation of Test plan with focus oTest Responsibilities, Prepare various Test Cases, Tabulate Test Results Responsibilities, Prepare various Test Cases, Tabulate Test Results	
12.	24-10-23	3 pm	5 pm	2 hours	Results Record the results Capture	

Name & Signature of Project Guide

Prof. S.B.Shirsat